

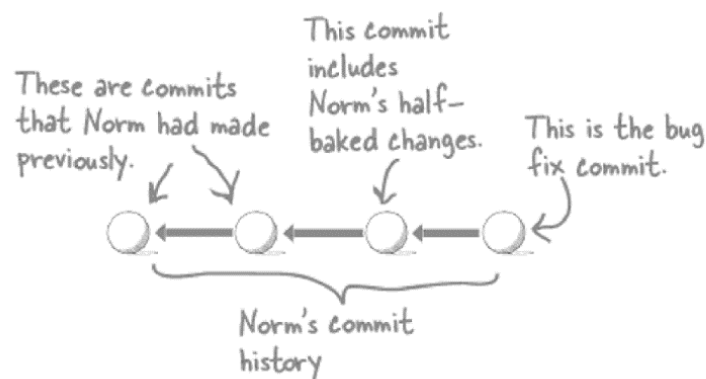
In-Class Exercise Sheet 4

1. Text and images from the book “Head First git” by Raju Gandh (O’Reilly).

Yesterday at work: Norman was so close to finishing up a complex change to the codebase that he could almost taste it.



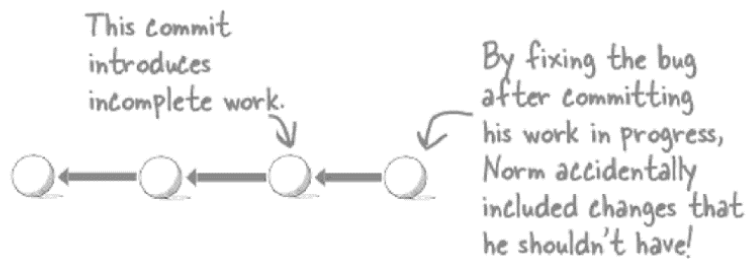
Norm knew he wasn’t done yet. But he committed his code anyway, and he started to tackle the bug. At the end of a long day, when he knew he had fixed that bug once and for all, he committed his work. This is what his commit history looked like now:



His project manager is not amused:



What could Norm have done different?



2. Code from the supplementary material from “Heads first git”, chapter 3.

You run `git log -p`:

```

1  commit 32367d76530da8fe77922aa905931ccf1fbd7524
2  Author: Rosemary Berry <headchef@80s-diner.com>
3  Date:   Wed Mar 17 10:28:12 2021 -0400
4
5      add punch
6
7  diff --git a/saucy.md b/saucy.md
8  index 20b7e5a..8d49c34 100644
9  --- a/saucy.md
10 +++ b/saucy.md
11 @@ -6,8 +6,8 @@
12  2 cups - Chopped cilantro
13  1/4 cup - Olive oil
14  1/4 cup - Lime juice
15  -1 pinch - Salt
16  -1 - Jalapeno, deseeded
17  +2 pinches - Salt
18  +2 - Jalapenos, deseeded
19
20  ## Instructions
21  -Add all ingredients to a blender. Mix until smooth.
22  +Add all ingredients to a blender. Mix until desired consistency.
23
24  commit 8d670e93e297b126fc0466caf37c0033b6c018ab
25  Author: Rosemary Berry <headchef@80s-diner.com>
26  Date:   Mon Mar 15 18:52:50 2021 -0400
27
28      update recipe name
29
30  diff --git a/saucy.md b/saucy.md
31  index 0b77f5b..20b7e5a 100644
32  --- a/saucy.md
33  +++ b/saucy.md
34  @@ -1,4 +1,4 @@
35  -# Call me Cilly
36  +# Spicy Green Mean Machine
37
38  ## Ingredients
39  1/2 cup - Plain yogurt
40
41  commit 4cca5a7b3225a88c2df12bd157f5877bc3dded9e
42  Author: Rosemary Berry <headchef@80s-diner.com>
43  Date:   Mon Feb 15 17:24:41 2021 -0500
44
45      make it spicy
46
47  diff --git a/saucy.md b/saucy.md
48  index affa4a4..0b77f5b 100644
49  --- a/saucy.md
50  +++ b/saucy.md
51  @@ -7,6 +7,7 @@
52  1/4 cup - Olive oil
53  1/4 cup - Lime juice
54  1 pinch - Salt
55  +1 - Jalapeno, deseeded
56
57  ## Instructions
58  Add all ingredients to a blender. Mix until smooth.
59
60  commit 5db2b68c4cf2699e89fe1b526a7e4a171d83b2ef
61  Author: Rosemary Berry <headchef@80s-diner.com>
62  Date:   Sat Jan 2 19:22:59 2021 -0500
63
64      first attempt
65
66  diff --git a/saucy.md b/saucy.md
67  new file mode 100644
68  index 0000000..affa4a4
69  --- /dev/null
70  +++ b/saucy.md
71  @@ -0,0 +1,12 @@
72  +# Call me Cilly
73  +
74  +## Ingredients
75  +1/2 cup - Plain yogurt
76  +3-4 cloves - Garlic
77  +2 cups - Chopped cilantro
78  +1/4 cup - Olive oil
79  +1/4 cup - Lime juice
80  +1 pinch - Salt
81  +
82  +## Instructions
83  +Add all ingredients to a blender. Mix until smooth.

```

What is the current content of the file `saucy.md`?

3. The figure below is from the publication “Beyond the Badge: Reproducibility Engineering as a Lifetime Skill” by W. Maurer, S. Klessinger, and S. Scherzinger, SEENG@ICSE 2022.

Below, we show a stylized software change, consisting of metadata and the technical change in form of a diff. Fill in the blanks to the right of the arrows, to explain how the patch is structured, and how the trail of responsibility is documented.

```
commit: aa09c4f6a54152... ← _____
Author: Jane Doe <jane@doe.com> ← _____
Committer: John Doe <john@doe.com> ← _____
```

```
Use salted hashes ← _____
Function getHash() is used to hash user passwords. Since adding a salt value is
considered a minimum standard these days, augment computing the hash with a
salting function as devised by Ilsebill et al., Grassian Letters 27(3), 2022.
```

```
Signed-off-by: Jane Doe <jane@doe.com> ← _____
Reviewed-by: Jean Doe <jean@doe.com> ← _____
Tested-by: Judy Doe <judy@doe.com> ← _____
diff -git a/sec/hash.c b/sec/hash.c ← _____
@@ -1,7 +1,7 @@
doSomething();
-hash = getHash(val);
+hash = getSaltedHash(val, salt()); ← _____
```

4. In your research, you build upon an open-source licensed git repository hosted on GitHub.

In building your reproduction package, what are the pros/cons between

- providing just the latest snapshot of the code, including your changes to the code,
- cloning the project, and then adding your code changes (e.g., as a git patch stack),
- forking the project, and independently working on your fork?

5. *The following content was developed in collaboration with Wolfgang Mauerer, OTH Regensburg, during an earlier iteration of this course.*

Below, we show the output of `git log` on the repository on <https://github.com/ReproEng/LabSession2>. The history documents the thought process during development. Now that development is complete, how could you restructure this git history so that the presentation is better structured and the history becomes more readable? For now, let's focus on the history as we *would want* to present it.

```

1  commit 009230fdb2136689f6a7dd7259a89cd64caf92c
2  Author: Wolfgang Mauerer <wolfgang.mauerer@othr.de>
3  Date:   Mon Oct 18 00:34:20 2021 +0200
4
5      fixup: Ensure that build system sets highest standards
6
7  diff --git a/Makefile b/Makefile
8  index 51e9c1d..0764d3c 100644
9  --- a/Makefile
10 +++ b/Makefile
11 @@ -1,5 +1,5 @@
12  .PHONY: clean
13  -CFLAGS=-w
14  +CFLAGS=-Wall -Werror -pedantic
15  CC=gcc
16
17  hello: hello.c
18
19  commit ab1f4d7696998f642d2a42fd2ded9ac7665b0856
20  Author: Wolfgang Mauerer <wolfgang.mauerer@othr.de>
21  Date:   Mon Oct 18 00:33:49 2021 +0200
22
23      fixup: Actually improve code quality
24
25  diff --git a/hello.c b/hello.c
26  index 8f4dad5..ed98a74 100644
27  --- a/hello.c
28  +++ b/hello.c
29 @@ -1,5 +1,6 @@
30  #include <stdio.h>
31
32  -void main() {
33  +int main() {
34      printf("Hello, world\n");
35  + return 0;
36  }
37
38  commit 8cdb59272635d10b182eb088ac84867b158a538e
39  Author: Wolfgang Mauerer <wolfgang.mauerer@othr.de>
40  Date:   Mon Oct 18 00:23:09 2021 +0200
41
42      Add build infrastructure
43
44      Out project is supposed to satisfy highest standards,
45      so make sure we automate the build process.
46
47      Signed-off-by: Wolfgang Mauerer <wolfgang.mauerer@othr.de>
48
49  diff --git a/.gitignore b/.gitignore
50  new file mode 100644
51  index 0000000..e0920f7
52  --- /dev/null
53  +++ b/.gitignore
54 @@ -0,0 +1,3 @@
55  +hello
56  +.*
57  +
58  diff --git a/Makefile b/Makefile
59  new file mode 100644
60  index 0000000..51e9c1d
61  --- /dev/null
62  +++ b/Makefile
63 @@ -0,0 +1,8 @@
64  +.PHONY: clean
65  +CFLAGS=-w
66  +CC=gcc
67  +
68  +hello: hello.c
69  +
70  +clean:
71  +    rm hello
72
73  commit d7fc97a96b53e9be1e3df19a80248c2e91100e37
74  Author: Wolfgang Mauerer <wolfgang.mauerer@othr.de>
75  Date:   Mon Oct 18 00:26:38 2021 +0200
76
77      Add code proper
78
79      An idea is nothing without a concrete implementation. Add
80      the actual program to the repository.
81
82      Signed-off-by: Wolfgang Mauerer <wolfgang.mauerer@othr.de>
83      Signed-off-by: Stefanie Scherzinger <stefanie.scherzinger@uni-passau.de>
84
85  diff --git a/hello.c b/hello.c
86  new file mode 100644
87  index 0000000..8f4dad5
88  --- /dev/null
89  +++ b/hello.c
90 @@ -0,0 +1,5 @@
91  +#include <stdio.h>
92  +
93  +void main() {
94  +    printf("Hello, world\n");
95  +}
96
97  commit 9cba6f7420fc4fe840abc347b33ea6aabb9e5032
98  Author: Wolfgang Mauerer <wolfgang.mauerer@othr.de>
99  Date:   Mon Oct 18 00:03:26 2021 +0200
100
101      Kick-Off a new project: The ultimate hello world tool
102
103      Since the sun rises every day, it seems appropriate
104      to greet the world in similar frequency. This important
105      task deserves a perfect implementation.
106
107      Signed-off-by: Wolfgang Mauerer <wolfgang.mauerer@othr.de>
108
109  diff --git a/README b/README
110  new file mode 100644
111  index 0000000..9664b45
112  --- /dev/null
113  +++ b/README
114 @@ -0,0 +1 @@
115  +In this repository, we will develop the ultimate hello world program.
116

```